

# **Simulation Analysis of Ohm's and Kirchhoff's Laws**

## **Model Properties and Design Description**

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## **Simulation Analysis of Ohm's and Kirchhoff's Laws: Model Properties and Design Description**

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# Chapter 1. Model Version

**Version:** 1.71

**Last modified:** Tue May 05 09:11:59 2026

**Checksum:** 3961229817 2706492340 3375769721 4022883664

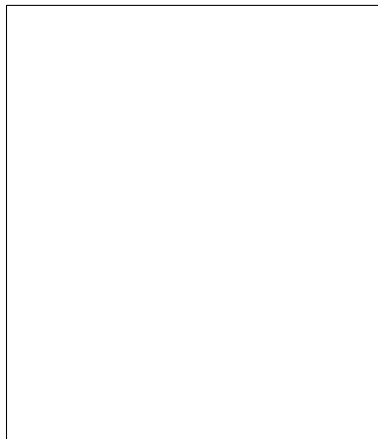
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# Chapter 2. Root System

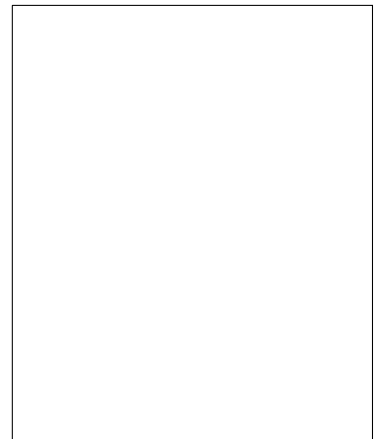
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**Figure 2.1. model\_tugas3**



Rangkaian Soal A



Rangkaian Soal B

## 2.1. Blocks

### 2.1.1. Parameters

### 2.1.2. Block Execution Order

1. EXEC\_STATE\_1 (SimscapeExecutionBlock)
2. EXEC\_OUTPUT\_3 (SimscapeExecutionBlock)
3. reshape (Reshape)
4. Arus\_I1(a) [5] (ToWorkspace)
5. Output I1A [8] (Display)
6. input (PMIOPort)
7. reshape (Reshape)
8. Arus\_I2(a) [6] (ToWorkspace)
9. Output I2A [9] (Display)
10. input (PMIOPort)
11. reshape (Reshape)
12. Arus\_I4(a) [6] (ToWorkspace)
13. Output I4A [9] (Display)
14. input (PMIOPort)
15. reshape (Reshape)



16. Output V1A [9] (Display)
17. V1A [12] (ToWorkspace)
18. input (PMIOPort)
19. reshape (Reshape)
20. Output V2A [9] (Display)
21. V2A [12] (ToWorkspace)
22. input (PMIOPort)
23. reshape (Reshape)
24. Output V3A [10] (Display)
25. V3A [13] (ToWorkspace)
26. input (PMIOPort)
27. reshape (Reshape)
28. Output V4A [10] (Display)
29. V4A [13] (ToWorkspace)
30. input (PMIOPort)
31. EXEC\_SINK\_2 (SimscapeExecutionBlock)
32. a (PMIOPort)
33. Current Sensor i1a [6] (SimscapeBlock)
34. Current Sensor i2a [7] (SimscapeBlock)
35. Current Sensor i4a [7] (SimscapeBlock)
36. DC Voltage Source [8] (SimscapeBlock)
37. Electrical Reference [8] (SimscapeBlock)
38. Ra1 [10] (SimscapeBlock)
39. Ra2 [11] (SimscapeBlock)
40. Ra3 [11] (SimscapeBlock)
41. Ra4 [11] (SimscapeBlock)
42. Voltage Sensor V1A [13] (SimscapeBlock)
43. Voltage Sensor V2A [14] (SimscapeBlock)
44. Voltage Sensor V3A [14] (SimscapeBlock)
45. Voltage Sensor V4A [15] (SimscapeBlock)
46. EXEC\_STATE\_1 (SimscapeExecutionBlock)
47. EXEC\_OUTPUT\_3 (SimscapeExecutionBlock)
48. reshape (Reshape)
49. Arus\_I1(b) [16] (ToWorkspace)
50. Output i1b(arus total) [19] (Display)
51. input (PMIOPort)
52. reshape (Reshape)
53. Arus\_I2(b) [16] (ToWorkspace)
54. Output i2b(arus cabang atas) [19] (Display)
55. input (PMIOPort)
56. reshape (Reshape)
57. Arus\_I4(b) [16] (ToWorkspace)
58. Output i4b(arus cabang bawah) [19] (Display)
59. input (PMIOPort)
60. reshape (Reshape)
61. Output v1b [19] (Display)
62. v1b [22] (ToWorkspace)
63. input (PMIOPort)
64. reshape (Reshape)
65. Output v2b [20] (Display)
66. v2b [23] (ToWorkspace)
67. input (PMIOPort)
68. reshape (Reshape)
69. Output v3b [20] (Display)

- 70. v3b [23] (ToWorkspace)
- 71. input (PMIOPort)
- 72. reshape (Reshape)
- 73. Output v4b [20] (Display)
- 74. v4b [23] (ToWorkspace)
- 75. input (PMIOPort)
- 76. EXEC\_SINK\_2 (SimscapeExecutionBlock)
- 77. a (PMIOPort)
- 78. Current Sensor i1b [17] (SimscapeBlock)
- 79. Current Sensor i2b [17] (SimscapeBlock)
- 80. Current Sensor i4B [17] (SimscapeBlock)
- 81. DC Voltage Source2 [18] (SimscapeBlock)
- 82. Electrical Reference1 [18] (SimscapeBlock)
- 83. Rb1 [20] (SimscapeBlock)
- 84. Rb2 [21] (SimscapeBlock)
- 85. Rb3 [21] (SimscapeBlock)
- 86. Rb4 [22] (SimscapeBlock)
- 87. Voltage Sensor v1b [24] (SimscapeBlock)
- 88. Voltage Sensor v2b [24] (SimscapeBlock)
- 89. Voltage Sensor v3b [25] (SimscapeBlock)
- 90. Voltage Sensor v4b [25] (SimscapeBlock)

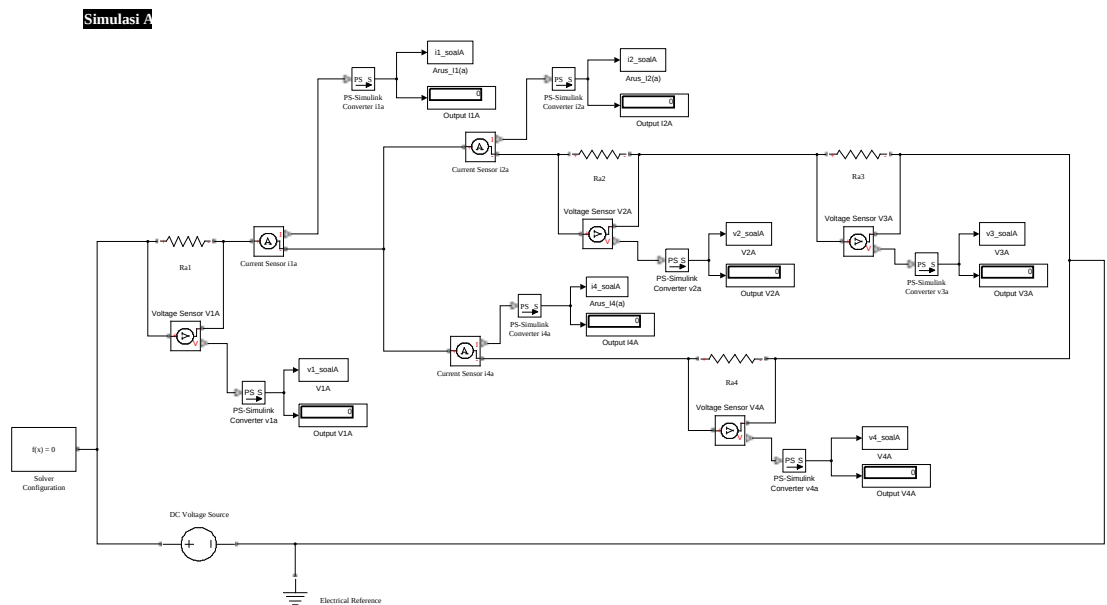
# Chapter 3. Subsystems

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## 3.1. Rangkaian Soal A

Figure 3.1. model\_tugas3/Rangkaian Soal A



### 3.1.1. Blocks

#### 3.1.1.1. Parameters

##### 3.1.1.1.1. "Arus\_I1(a)" (ToWorkspace)

Table 3.1. "Arus\_I1(a)" Parameters

| Parameter                 | Value    |
|---------------------------|----------|
| Variable name             | i1_soalA |
| Limit data points to last | inf      |
| Decimation                | 1        |

| Parameter                           | Value |
|-------------------------------------|-------|
| Sample time (-1 for inherited)      | -1    |
| Save format                         | Array |
| Log fixed-point data as a fi object | off   |

### 3.1.1.1.2. "Arus\_I2(a)" (ToWorkspace)

**Table 3.2. "Arus\_I2(a)" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | i2_soalA |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

### 3.1.1.1.3. "Arus\_I4(a)" (ToWorkspace)

**Table 3.3. "Arus\_I4(a)" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | i4_soalA |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

### 3.1.1.1.4. "Current Sensor i1a" (SimscapeBlock)

**Table 3.4. "Current Sensor i1a" Parameters**

| Parameter         | Value                                 |
|-------------------|---------------------------------------|
| ComponentPath     | foundation.electrical.sensors.current |
| ComponentVariants | foundation.electrical.sensors.current |

| Parameter             | Value   |
|-----------------------|---------|
| ComponentVariantNames | current |
| ClassName             | current |
| SchemaVersion         | 1       |
| i1_Log                | off     |
| v1_Log                | off     |
| LocalVarNames         | i1 v1   |
| LocalVarDescs         | i1 v1   |
| LocalVarLogging       | [0 0]   |

### 3.1.1.1.5. "Current Sensor i2a" (SimscapeBlock)

**Table 3.5. "Current Sensor i2a" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.current |
| ComponentVariants     | foundation.electrical.sensors.current |
| ComponentVariantNames | current                               |
| ClassName             | current                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

### 3.1.1.1.6. "Current Sensor i4a" (SimscapeBlock)

**Table 3.6. "Current Sensor i4a" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.current |
| ComponentVariants     | foundation.electrical.sensors.current |
| ComponentVariantNames | current                               |
| ClassName             | current                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |

| Parameter       | Value |
|-----------------|-------|
| LocalVarNames   | i1 v1 |
| LocalVarDescs   | i1 v1 |
| LocalVarLogging | [0 0] |

### 3.1.1.1.7. "DC Voltage Source" (SimscapeBlock)

**Table 3.7. "DC Voltage Source" Parameters**

| Parameter             | Value                                    |
|-----------------------|------------------------------------------|
| ComponentPath         | foundation.electrical.sources.dc_voltage |
| ComponentVariants     | foundation.electrical.sources.dc_voltage |
| ComponentVariantNames | dc_voltage                               |
| ClassName             | dc_voltage                               |
| SchemaVersion         | 1                                        |
| Constant voltage      | V                                        |
| v0_unit               | V                                        |
| i_Log                 | off                                      |
| v_Log                 | off                                      |
| LocalVarNames         | i v                                      |
| LocalVarDescs         | i v                                      |
| LocalVarLogging       | [0 0]                                    |

### 3.1.1.1.8. "Electrical Reference" (SimscapeBlock)

**Table 3.8. "Electrical Reference" Parameters**

| Parameter             | Value                                    |
|-----------------------|------------------------------------------|
| ComponentPath         | foundation.electrical.elements.reference |
| ComponentVariants     | foundation.electrical.elements.reference |
| ComponentVariantNames | reference                                |
| ClassName             | reference                                |
| SchemaVersion         | 1                                        |
| i_Log                 | off                                      |
| LocalVarNames         | i                                        |
| LocalVarDescs         | i                                        |
| LocalVarLogging       | 0                                        |

**3.1.1.1.9. "Output I1A" (Display)****Table 3.9. "Output I1A" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.1.1.1.10. "Output I2A" (Display)****Table 3.10. "Output I2A" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.1.1.1.11. "Output I4A" (Display)****Table 3.11. "Output I4A" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.1.1.1.12. "Output V1A" (Display)****Table 3.12. "Output V1A" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.1.1.1.13. "Output V2A" (Display)****Table 3.13. "Output V2A" Parameters**

| Parameter | Value |
|-----------|-------|
| Format    | short |

| Parameter        | Value |
|------------------|-------|
| Decimation       | 1     |
| Floating display | off   |

#### 3.1.1.1.14. "Output V3A" (Display)

**Table 3.14. "Output V3A" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

#### 3.1.1.1.15. "Output V4A" (Display)

**Table 3.15. "Output V4A" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

#### 3.1.1.1.16. "Ra1" (SimscapeBlock)

**Table 3.16. "Ra1" Parameters**

| Parameter             | Value                                   |
|-----------------------|-----------------------------------------|
| ComponentPath         | foundation.electrical.elements.resistor |
| ComponentVariants     | foundation.electrical.elements.resistor |
| ComponentVariantNames | resistor                                |
| ClassName             | resistor                                |
| SchemaVersion         | 1                                       |
| Resistance            | R1                                      |
| R_unit                | Ohm                                     |
| i_Log                 | off                                     |
| v_Log                 | off                                     |
| LocalVarNames         | i v                                     |
| LocalVarDescs         | i v                                     |
| LocalVarLogging       | [0 0]                                   |



**3.1.1.1.17. "Ra2" (SimscapeBlock)****Table 3.17. "Ra2" Parameters**

| Parameter             | Value                                   |
|-----------------------|-----------------------------------------|
| ComponentPath         | foundation.electrical.elements.resistor |
| ComponentVariants     | foundation.electrical.elements.resistor |
| ComponentVariantNames | resistor                                |
| ClassName             | resistor                                |
| SchemaVersion         | 1                                       |
| Resistance            | R2                                      |
| R_unit                | Ohm                                     |
| i_Log                 | off                                     |
| v_Log                 | off                                     |
| LocalVarNames         | i v                                     |
| LocalVarDescs         | i v                                     |
| LocalVarLogging       | [0 0]                                   |

**3.1.1.1.18. "Ra3" (SimscapeBlock)****Table 3.18. "Ra3" Parameters**

| Parameter             | Value                                   |
|-----------------------|-----------------------------------------|
| ComponentPath         | foundation.electrical.elements.resistor |
| ComponentVariants     | foundation.electrical.elements.resistor |
| ComponentVariantNames | resistor                                |
| ClassName             | resistor                                |
| SchemaVersion         | 1                                       |
| Resistance            | R3                                      |
| R_unit                | Ohm                                     |
| i_Log                 | off                                     |
| v_Log                 | off                                     |
| LocalVarNames         | i v                                     |
| LocalVarDescs         | i v                                     |
| LocalVarLogging       | [0 0]                                   |

**3.1.1.1.19. "Ra4" (SimscapeBlock)****Table 3.19. "Ra4" Parameters**

| Parameter             | Value                                   |
|-----------------------|-----------------------------------------|
| ComponentPath         | foundation.electrical.elements.resistor |
| ComponentVariants     | foundation.electrical.elements.resistor |
| ComponentVariantNames | resistor                                |
| ClassName             | resistor                                |
| SchemaVersion         | 1                                       |
| Resistance            | R4                                      |
| R_unit                | Ohm                                     |
| i_Log                 | off                                     |
| v_Log                 | off                                     |
| LocalVarNames         | i v                                     |
| LocalVarDescs         | i v                                     |
| LocalVarLogging       | [0 0]                                   |

**3.1.1.1.20. "V1A" (ToWorkspace)****Table 3.20. "V1A" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | v1_soalA |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

**3.1.1.1.21. "V2A" (ToWorkspace)****Table 3.21. "V2A" Parameters**

| Parameter                      | Value    |
|--------------------------------|----------|
| Variable name                  | v2_soalA |
| Limit data points to last      | inf      |
| Decimation                     | 1        |
| Sample time (-1 for inherited) | -1       |

| Parameter                           | Value |
|-------------------------------------|-------|
| Save format                         | Array |
| Log fixed-point data as a fi object | off   |

### 3.1.1.1.22. "V3A" (ToWorkspace)

**Table 3.22. "V3A" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | v3_soalA |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

### 3.1.1.1.23. "V4A" (ToWorkspace)

**Table 3.23. "V4A" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | v4_soalA |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

### 3.1.1.1.24. "Voltage Sensor V1A" (SimscapeBlock)

**Table 3.24. "Voltage Sensor V1A" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.voltage |
| ComponentVariants     | foundation.electrical.sensors.voltage |
| ComponentVariantNames | voltage                               |

| Parameter       | Value   |
|-----------------|---------|
| ClassName       | voltage |
| SchemaVersion   | 1       |
| i1_Log          | off     |
| v1_Log          | off     |
| LocalVarNames   | i1 v1   |
| LocalVarDescs   | i1 v1   |
| LocalVarLogging | [0 0]   |

### 3.1.1.1.25. "Voltage Sensor V2A" (SimscapeBlock)

**Table 3.25. "Voltage Sensor V2A" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.voltage |
| ComponentVariants     | foundation.electrical.sensors.voltage |
| ComponentVariantNames | voltage                               |
| ClassName             | voltage                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

### 3.1.1.1.26. "Voltage Sensor V3A" (SimscapeBlock)

**Table 3.26. "Voltage Sensor V3A" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.voltage |
| ComponentVariants     | foundation.electrical.sensors.voltage |
| ComponentVariantNames | voltage                               |
| ClassName             | voltage                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |

| Parameter       | Value |
|-----------------|-------|
| LocalVarLogging | [0 0] |

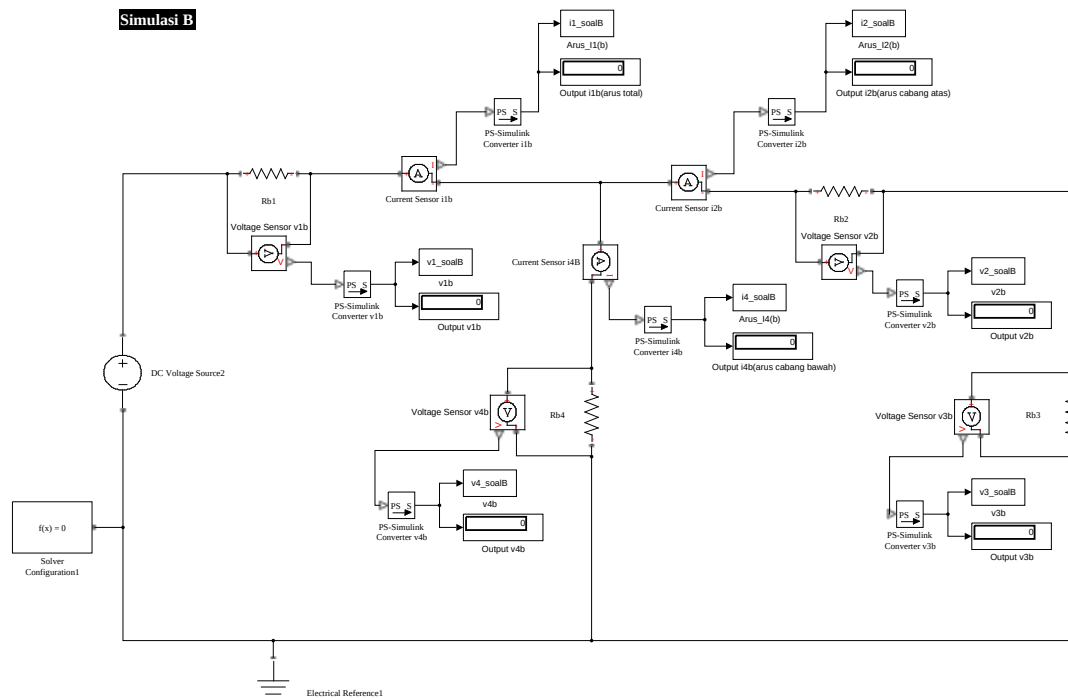
### 3.1.1.1.27. "Voltage Sensor V4A" (SimscapeBlock)

**Table 3.27. "Voltage Sensor V4A" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.voltage |
| ComponentVariants     | foundation.electrical.sensors.voltage |
| ComponentVariantNames | voltage                               |
| ClassName             | voltage                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

## 3.2. Rangkaian Soal B

**Figure 3.2. model\_tugas3/Rangkaian Soal B**



## 3.2.1. Blocks

### 3.2.1.1. Parameters

#### 3.2.1.1.1. "Arus\_I1(b)" (ToWorkspace)

**Table 3.28. "Arus\_I1(b)" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | i1_soalB |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

#### 3.2.1.1.2. "Arus\_I2(b)" (ToWorkspace)

**Table 3.29. "Arus\_I2(b)" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | i2_soalB |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

#### 3.2.1.1.3. "Arus\_I4(b)" (ToWorkspace)

**Table 3.30. "Arus\_I4(b)" Parameters**

| Parameter                      | Value    |
|--------------------------------|----------|
| Variable name                  | i4_soalB |
| Limit data points to last      | inf      |
| Decimation                     | 1        |
| Sample time (-1 for inherited) | -1       |

| Parameter                           | Value |
|-------------------------------------|-------|
| Save format                         | Array |
| Log fixed-point data as a fi object | off   |

#### 3.2.1.1.4. "Current Sensor i1b" (SimscapeBlock)

**Table 3.31. "Current Sensor i1b" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.current |
| ComponentVariants     | foundation.electrical.sensors.current |
| ComponentVariantNames | current                               |
| ClassName             | current                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

#### 3.2.1.1.5. "Current Sensor i2b" (SimscapeBlock)

**Table 3.32. "Current Sensor i2b" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.current |
| ComponentVariants     | foundation.electrical.sensors.current |
| ComponentVariantNames | current                               |
| ClassName             | current                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

### 3.2.1.1.6. "Current Sensor i4B" (SimscapeBlock)

**Table 3.33. "Current Sensor i4B" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.current |
| ComponentVariants     | foundation.electrical.sensors.current |
| ComponentVariantNames | current                               |
| ClassName             | current                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

### 3.2.1.1.7. "DC Voltage Source2" (SimscapeBlock)

**Table 3.34. "DC Voltage Source2" Parameters**

| Parameter             | Value                                    |
|-----------------------|------------------------------------------|
| ComponentPath         | foundation.electrical.sources.dc_voltage |
| ComponentVariants     | foundation.electrical.sources.dc_voltage |
| ComponentVariantNames | dc_voltage                               |
| ClassName             | dc_voltage                               |
| SchemaVersion         | 1                                        |
| Constant voltage      | V                                        |
| v0_unit               | V                                        |
| i_Log                 | off                                      |
| v_Log                 | off                                      |
| LocalVarNames         | i v                                      |
| LocalVarDescs         | i v                                      |
| LocalVarLogging       | [0 0]                                    |

### 3.2.1.1.8. "Electrical Reference1" (SimscapeBlock)

**Table 3.35. "Electrical Reference1" Parameters**

| Parameter     | Value                                    |
|---------------|------------------------------------------|
| ComponentPath | foundation.electrical.elements.reference |



| Parameter             | Value                                    |
|-----------------------|------------------------------------------|
| ComponentVariants     | foundation.electrical.elements.reference |
| ComponentVariantNames | reference                                |
| ClassName             | reference                                |
| SchemaVersion         | 1                                        |
| i_Log                 | off                                      |
| LocalVarNames         | i                                        |
| LocalVarDescs         | i                                        |
| LocalVarLogging       | 0                                        |

### 3.2.1.1.9. "Output i1b(arus total)" (Display)

**Table 3.36. "Output i1b(arus total)" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

### 3.2.1.1.10. "Output i2b(arus cabang atas)" (Display)

**Table 3.37. "Output i2b(arus cabang atas)" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

### 3.2.1.1.11. "Output i4b(arus cabang bawah)" (Display)

**Table 3.38. "Output i4b(arus cabang bawah)" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.2.1.1.12. "Output v1b" (Display)****Table 3.39. "Output v1b" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.2.1.1.13. "Output v2b" (Display)****Table 3.40. "Output v2b" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.2.1.1.14. "Output v3b" (Display)****Table 3.41. "Output v3b" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.2.1.1.15. "Output v4b" (Display)****Table 3.42. "Output v4b" Parameters**

| Parameter        | Value |
|------------------|-------|
| Format           | short |
| Decimation       | 1     |
| Floating display | off   |

**3.2.1.1.16. "Rb1" (SimscapeBlock)****Table 3.43. "Rb1" Parameters**

| Parameter     | Value                                   |
|---------------|-----------------------------------------|
| ComponentPath | foundation.electrical.elements.resistor |

| Parameter             | Value                                   |
|-----------------------|-----------------------------------------|
| ComponentVariants     | foundation.electrical.elements.resistor |
| ComponentVariantNames | resistor                                |
| ClassName             | resistor                                |
| SchemaVersion         | 1                                       |
| Resistance            | R1                                      |
| R_unit                | Ohm                                     |
| i_Log                 | off                                     |
| v_Log                 | off                                     |
| LocalVarNames         | i v                                     |
| LocalVarDescs         | i v                                     |
| LocalVarLogging       | [0 0]                                   |

### 3.2.1.1.17. "Rb2" (SimscapeBlock)

**Table 3.44. "Rb2" Parameters**

| Parameter             | Value                                   |
|-----------------------|-----------------------------------------|
| ComponentPath         | foundation.electrical.elements.resistor |
| ComponentVariants     | foundation.electrical.elements.resistor |
| ComponentVariantNames | resistor                                |
| ClassName             | resistor                                |
| SchemaVersion         | 1                                       |
| Resistance            | R2                                      |
| R_unit                | Ohm                                     |
| i_Log                 | off                                     |
| v_Log                 | off                                     |
| LocalVarNames         | i v                                     |
| LocalVarDescs         | i v                                     |
| LocalVarLogging       | [0 0]                                   |

### 3.2.1.1.18. "Rb3" (SimscapeBlock)

**Table 3.45. "Rb3" Parameters**

| Parameter         | Value                                   |
|-------------------|-----------------------------------------|
| ComponentPath     | foundation.electrical.elements.resistor |
| ComponentVariants | foundation.electrical.elements.resistor |

| Parameter             | Value    |
|-----------------------|----------|
| ComponentVariantNames | resistor |
| ClassName             | resistor |
| SchemaVersion         | 1        |
| Resistance            | R3       |
| R_unit                | Ohm      |
| i_Log                 | off      |
| v_Log                 | off      |
| LocalVarNames         | i v      |
| LocalVarDescs         | i v      |
| LocalVarLogging       | [0 0]    |

### 3.2.1.1.19. "Rb4" (SimscapeBlock)

**Table 3.46. "Rb4" Parameters**

| Parameter             | Value                                   |
|-----------------------|-----------------------------------------|
| ComponentPath         | foundation.electrical.elements.resistor |
| ComponentVariants     | foundation.electrical.elements.resistor |
| ComponentVariantNames | resistor                                |
| ClassName             | resistor                                |
| SchemaVersion         | 1                                       |
| Resistance            | R4                                      |
| R_unit                | Ohm                                     |
| i_Log                 | off                                     |
| v_Log                 | off                                     |
| LocalVarNames         | i v                                     |
| LocalVarDescs         | i v                                     |
| LocalVarLogging       | [0 0]                                   |

### 3.2.1.1.20. "v1b" (ToWorkspace)

**Table 3.47. "v1b" Parameters**

| Parameter                 | Value    |
|---------------------------|----------|
| Variable name             | v1_soalB |
| Limit data points to last | inf      |
| Decimation                | 1        |

| Parameter                           | Value |
|-------------------------------------|-------|
| Sample time (-1 for inherited)      | -1    |
| Save format                         | Array |
| Log fixed-point data as a fi object | off   |

### 3.2.1.1.21. "v2b" (ToWorkspace)

**Table 3.48. "v2b" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | v2_soalB |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

### 3.2.1.1.22. "v3b" (ToWorkspace)

**Table 3.49. "v3b" Parameters**

| Parameter                           | Value    |
|-------------------------------------|----------|
| Variable name                       | v3_soalB |
| Limit data points to last           | inf      |
| Decimation                          | 1        |
| Sample time (-1 for inherited)      | -1       |
| Save format                         | Array    |
| Log fixed-point data as a fi object | off      |

### 3.2.1.1.23. "v4b" (ToWorkspace)

**Table 3.50. "v4b" Parameters**

| Parameter     | Value    |
|---------------|----------|
| Variable name | v4_soalB |

| Parameter                           | Value |
|-------------------------------------|-------|
| Limit data points to last           | inf   |
| Decimation                          | 1     |
| Sample time (-1 for inherited)      | -1    |
| Save format                         | Array |
| Log fixed-point data as a fi object | off   |

### 3.2.1.1.24. "Voltage Sensor v1b" (SimscapeBlock)

**Table 3.51. "Voltage Sensor v1b" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.voltage |
| ComponentVariants     | foundation.electrical.sensors.voltage |
| ComponentVariantNames | voltage                               |
| ClassName             | voltage                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

### 3.2.1.1.25. "Voltage Sensor v2b" (SimscapeBlock)

**Table 3.52. "Voltage Sensor v2b" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.voltage |
| ComponentVariants     | foundation.electrical.sensors.voltage |
| ComponentVariantNames | voltage                               |
| ClassName             | voltage                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |

| Parameter       | Value |
|-----------------|-------|
| LocalVarLogging | [0 0] |

### 3.2.1.1.26. "Voltage Sensor v3b" (SimscapeBlock)

**Table 3.53. "Voltage Sensor v3b" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.voltage |
| ComponentVariants     | foundation.electrical.sensors.voltage |
| ComponentVariantNames | voltage                               |
| ClassName             | voltage                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

### 3.2.1.1.27. "Voltage Sensor v4b" (SimscapeBlock)

**Table 3.54. "Voltage Sensor v4b" Parameters**

| Parameter             | Value                                 |
|-----------------------|---------------------------------------|
| ComponentPath         | foundation.electrical.sensors.voltage |
| ComponentVariants     | foundation.electrical.sensors.voltage |
| ComponentVariantNames | voltage                               |
| ClassName             | voltage                               |
| SchemaVersion         | 1                                     |
| i1_Log                | off                                   |
| v1_Log                | off                                   |
| LocalVarNames         | i1 v1                                 |
| LocalVarDescs         | i1 v1                                 |
| LocalVarLogging       | [0 0]                                 |

---

# Chapter 4. System Design Variables

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## 4.1. Design Variable Summary

**Table 4.1. Design Variables**

| Variable Name | Parent Blocks                                    | Size | Bytes | Class  | Value |
|---------------|--------------------------------------------------|------|-------|--------|-------|
| R1            | Ra1 [10]<br>Rb1 [20]                             | 1x1  | 8     | double | 10    |
| R2            | Ra2 [11]<br>Rb2 [21]                             | 1x1  | 8     | double | 10    |
| R3            | Ra3 [11]<br>Rb3 [21]                             | 1x1  | 8     | double | 10    |
| R4            | Ra4 [11]<br>Rb4 [22]                             | 1x1  | 8     | double | 10    |
| V             | DC Voltage Source [8]<br>DC Voltage Source2 [18] | 1x1  | 8     | double | 9     |

## 4.2. Design Variable Details

**R1. 10**

**Used by Blocks:**

- model\_tugas3/Rangkaian Soal A/Ra1 [10]
- model\_tugas3/Rangkaian Soal B/Rb1 [20]
- model\_tugas3/Rangkaian Soal A/Ra1 [10]
- model\_tugas3/Rangkaian Soal B/Rb1 [20]

Workspace: model (model\_tugas3)

**R2. 10**

**Used by Blocks:**

- model\_tugas3/Rangkaian Soal A/Ra2 [11]
- model\_tugas3/Rangkaian Soal B/Rb2 [21]
- model\_tugas3/Rangkaian Soal A/Ra2 [11]
- model\_tugas3/Rangkaian Soal B/Rb2 [21]



Workspace: model (model\_tugas3)

**R3. 10**

**Used by Blocks:**

- model\_tugas3/Rangkaian Soal A/Ra3 [11]
- model\_tugas3/Rangkaian Soal B/Rb3 [21]
- model\_tugas3/Rangkaian Soal A/Ra3 [11]
- model\_tugas3/Rangkaian Soal B/Rb3 [21]

Workspace: model (model\_tugas3)

**R4. 10**

**Used by Blocks:**

- model\_tugas3/Rangkaian Soal A/Ra4 [11]
- model\_tugas3/Rangkaian Soal B/Rb4 [22]
- model\_tugas3/Rangkaian Soal A/Ra4 [11]
- model\_tugas3/Rangkaian Soal B/Rb4 [22]

Workspace: model (model\_tugas3)

**V. 9**

**Used by Blocks:**

- model\_tugas3/Rangkaian Soal A/DC Voltage Source [8]
- model\_tugas3/Rangkaian Soal B/DC Voltage Source2 [18]
- model\_tugas3/Rangkaian Soal A/DC Voltage Source [8]
- model\_tugas3/Rangkaian Soal B/DC Voltage Source2 [18]

Workspace: model (model\_tugas3)

---

# Chapter 5. Requirements Traceability

model\_tugas3 does not contain requirements traceability links.

---

# Chapter 6. System Model Configuration

**Table 6.1. model\_tugas3 Configuration Set**

| Property       | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Components     | [ model_tugas3 Configuration Set.Components(1)-[29], model_tugas3 Configuration Set.Components(2) [30], model_tugas3 Configuration Set.Components(3) [31], model_tugas3 Configuration Set.Components(4) [32], model_tugas3 Configuration Set.Components(5) [34], model_tugas3 Configuration Set.Components(6-)[35], model_tugas3 Configuration Set.Components(7) [36], model_tugas3 Configuration Set.Components(8) [37], model_tugas3 Configuration Set.Components(9) [38], model_tugas3 Configuration Set.Components(10) [39] ] |
| Name           | Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| SimulationMode | normal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Table 6.2. model\_tugas3 Configuration Set.Components [29](1)**

| Property                 | Value      |
|--------------------------|------------|
| Name                     | Solver     |
| Description              |            |
| Components               |            |
| StartTime                | 0.0        |
| StopTime                 | 1.0        |
| AbsTol                   | auto       |
| FixedStep                | auto       |
| InitialStep              | auto       |
| MaxNumMinSteps           | -1         |
| MaxOrder                 | 5          |
| ZcThreshold              | auto       |
| ConsecutiveZCsStepRelTol | 10*128*eps |
| MaxConsecutiveZCs        | 1000       |
| ExtrapolationOrder       | 4          |
| NumberNewtonIterations   | 1          |
| MaxStep                  | auto       |
| MinStep                  | auto       |
| MaxConsecutiveMinStep    | 1          |
| RelTol                   | 1e-3       |

|                             |                   |
|-----------------------------|-------------------|
| SolverMode                  | Auto              |
| EnableConcurrentExecution   | off               |
| ConcurrentTasks             | off               |
| Solver                      | ode15s            |
| SolverName                  | ode15s            |
| SolverType                  | Variable-step     |
| SolverJacobianMethodControl | auto              |
| ShapePreserveControl        | DisableAll        |
| ZeroCrossControl            | UseLocalSettings  |
| ZeroCrossAlgorithm          | Nonadaptive       |
| SolverResetMethod           | Fast              |
| PositivePriorityOrder       | off               |
| AutoInsertRateTranBlk       | off               |
| SampleTimeConstraint        | Unconstrained     |
| InsertRTBMode               | Whenever possible |
| SampleTimeProperty          |                   |

**Table 6.3. model\_tugas3 Configuration Set.Components [29](2)**

| Property                  | Value              |
|---------------------------|--------------------|
| Name                      | Data Import/Export |
| Description               |                    |
| Components                |                    |
| Decimation                | 1                  |
| ExternalInput             | [t, u]             |
| FinalStateName            | xFinal             |
| InitialState              | xInitial           |
| LimitDataPoints           | on                 |
| MaxDataPoints             | 1000               |
| LoadExternalInput         | off                |
| LoadInitialState          | off                |
| SaveFinalState            | off                |
| SaveCompleteFinalSimState | off                |
| SaveFormat                | Array              |
| SignalLoggingSaveFormat   | Dataset            |
| SaveOutput                | on                 |
| SaveState                 | off                |
| SignalLogging             | on                 |
| DSMLogging                | on                 |
| InspectSignalLogs         | off                |

|                            |                   |
|----------------------------|-------------------|
| SaveTime                   | on                |
| ReturnWorkspaceOutputs     | off               |
| StateSaveName              | xout              |
| TimeSaveName               | tout              |
| OutputSaveName             | yout              |
| SignalLoggingName          | logsout           |
| DSMLoggingName             | dsmout            |
| OutputOption               | RefineOutputTimes |
| OutputTimes                | []                |
| ReturnWorkspaceOutputsName | out               |
| Refine                     | 1                 |

**Table 6.4. model\_tugas3 Configuration Set.Components [29](3)**

| Property                     | Value               |
|------------------------------|---------------------|
| Name                         | Optimization        |
| Description                  |                     |
| Components                   |                     |
| BlockReduction               | on                  |
| BooleanDataType              | on                  |
| ConditionallyExecuteInputs   | on                  |
| InlineParams                 | off                 |
| UseIntDivNetSlope            | off                 |
| UseFloatMulNetSlope          | off                 |
| UseSpecifiedMinMax           | off                 |
| InlineInvariantSignals       | off                 |
| OptimizeBlockIOStorage       | on                  |
| BufferReuse                  | on                  |
| EnhancedBackFolding          | off                 |
| StrengthReduction            | off                 |
| AdvancedOptControl           |                     |
| EnforceIntegerDowncast       | on                  |
| ExpressionFolding            | on                  |
| BooleansAsBitfields          | off                 |
| BitfieldContainerType        | uint_T              |
| EnableMemcpy                 | on                  |
| MemcpyThreshold              | 64                  |
| PassReuseOutputArgsAs        | Structure reference |
| PassReuseOutputArgsThreshold | 12                  |
| FoldNonRolledExpr            | on                  |

|                                     |                     |
|-------------------------------------|---------------------|
| LocalBlockOutputs                   | on                  |
| RollThreshold                       | 5                   |
| SystemCodeInlineAuto                | off                 |
| StateBitsets                        | off                 |
| DataBitsets                         | off                 |
| UseTempVars                         | off                 |
| ZeroExternalMemoryAtStartup         | on                  |
| ZeroInternalMemoryAtStartup         | on                  |
| InitFltsAndDblsToZero               | off                 |
| NoFixptDivByZeroProtection          | off                 |
| EfficientFloat2IntCast              | off                 |
| EfficientMapNaN2IntZero             | on                  |
| OptimizeModelRefInitCode            | off                 |
| LifeSpan                            | inf                 |
| EvaledLifeSpan                      | Inf                 |
| MaxStackSize                        | Inherit from target |
| BufferReusableBoundary              | on                  |
| SimCompilerOptimization             | Off                 |
| AccelVerboseBuild                   | off                 |
| ParallelExecutionInRapidAccelerator | on                  |

**Table 6.5. model\_tugas3 Configuration Set.Components [29](4)**

| Property                      | Value            |
|-------------------------------|------------------|
| Name                          | Diagnostics      |
| Description                   |                  |
| Components                    |                  |
| RTPrefix                      | error            |
| ConsistencyChecking           | none             |
| ArrayBoundsChecking           | none             |
| SignalInfNanChecking          | none             |
| SignalRangeChecking           | none             |
| ReadBeforeWriteMsg            | UseLocalSettings |
| WriteAfterWriteMsg            | UseLocalSettings |
| WriteAfterReadMsg             | UseLocalSettings |
| AlgebraicLoopMsg              | warning          |
| ArtificialAlgebraicLoopMsg    | warning          |
| SaveWithDisabledLinksMsg      | warning          |
| SaveWithParameterizedLinksMsg | warning          |
| CheckSSInitialOutputMsg       | on               |

|                                        |                  |
|----------------------------------------|------------------|
| UnderspecifiedInitializationDetection  | Classic          |
| MergeDetectMultiDrivingBlocksExec      | none             |
| CheckExecutionContextPreStartOutputMsg | off              |
| CheckExecutionContextRuntimeOutputMsg  | off              |
| SignalResolutionControl                | UseLocalSettings |
| BlockPriorityViolationMsg              | warning          |
| MinStepSizeMsg                         | warning          |
| TimeAdjustmentMsg                      | none             |
| MaxConsecutiveZCsMsg                   | error            |
| MaskedZcDiagnostic                     | warning          |
| IgnoredZcDiagnostic                    | warning          |
| SolverPrmCheckMsg                      | none             |
| InheritedTsInSrcMsg                    | warning          |
| DiscreteInheritContinuousMsg           | warning          |
| MultiTaskDSMMsg                        | error            |
| MultiTaskCondExecSysMsg                | error            |
| MultiTaskRateTransMsg                  | error            |
| SingleTaskRateTransMsg                 | none             |
| TasksWithSamePriorityMsg               | warning          |
| SigSpecEnsureSampleTimeMsg             | warning          |
| CheckMatrixSingularityMsg              | none             |
| IntegerOverflowMsg                     | warning          |
| Int32ToFloatConvMsg                    | warning          |
| ParameterDowncastMsg                   | error            |
| ParameterOverflowMsg                   | error            |
| ParameterUnderflowMsg                  | none             |
| ParameterPrecisionLossMsg              | warning          |
| ParameterTunabilityLossMsg             | warning          |
| FixptConstUnderflowMsg                 | none             |
| FixptConstOverflowMsg                  | none             |
| FixptConstPrecisionLossMsg             | none             |
| UnderSpecifiedDataTypeMsg              | none             |
| UnnecessaryDatatypeConvMsg             | none             |
| VectorMatrixConversionMsg              | none             |
| InvalidFcnCallConnMsg                  | error            |
| FcnCallInpInsideContextMsg             | EnableAllAsError |
| SignalLabelMismatchMsg                 | none             |
| UnconnectedInputMsg                    | warning          |
| UnconnectedOutputMsg                   | warning          |

|                                         |                  |
|-----------------------------------------|------------------|
| UnconnectedLineMsg                      | warning          |
| SFCnCompatibilityMsg                    | none             |
| FrameProcessingCompatibilityMsg         | warning          |
| UniqueDataStoreMsg                      | none             |
| BusObjectLabelMismatch                  | warning          |
| RootOutputRequireBusObject              | warning          |
| AssertControl                           | UseLocalSettings |
| Echo                                    |                  |
| EnableOverflowDetection                 | off              |
| ModelReferenceIOMsg                     | none             |
| ModelReferenceVersionMismatchMessage    | none             |
| ModelReferenceIOMismatchMessage         | none             |
| ModelReferenceCSMismatchMessage         | none             |
| ModelReferenceSimTargetVerbose          | off              |
| UnknownTsInhSupMsg                      | warning          |
| ModelReferenceDataLoggingMessage        | warning          |
| ModelReferenceSymbolNameMessage         | warning          |
| ModelReferenceExtraNoncontSigs          | error            |
| StateNameClashWarn                      | warning          |
| SimStateInterfaceChecksumMismatchMsg    | warning          |
| SimStateOlderReleaseMsg                 | error            |
| InitInArrayFormatMsg                    | warning          |
| StrictBusMsg                            | ErrorLevel1      |
| BusNameAdapt                            | WarnAndRepair    |
| NonBusSignalsTreatedAsBus               | none             |
| LoggingUnavailableSignals               | error            |
| SFUnusedDataAndEventsDiag               | warning          |
| SFUnexpectedBacktrackingDiag            | warning          |
| SFInvalidInputDataAccessInChartInitDiag | warning          |
| SFNoUnconditionalDefaultTransitionDiag  | warning          |
| SFTransitionOutsideNaturalParentDiag    | warning          |
| SFUnconditionalTransitionShadowingDiag  | warning          |
| SFUndirectedBroadcastEventsDiag         | warning          |
| SFTransitionActionBeforeConditionDiag   | warning          |

**Table 6.6. model\_tugas3 Configuration Set.Components [29](5)**

| Property    | Value                   |
|-------------|-------------------------|
| Name        | Hardware Implementation |
| Description |                         |



| Components                           |                |
|--------------------------------------|----------------|
| ProdBitPerChar                       | 8              |
| ProdBitPerShort                      | 16             |
| ProdBitPerInt                        | 32             |
| ProdBitPerLong                       | 32             |
| ProdBitPerLongLong                   | 64             |
| ProdBitPerFloat                      | 32             |
| ProdBitPerDouble                     | 64             |
| ProdBitPerPointer                    | 32             |
| ProdLargestAtomicInteger             | Char           |
| ProdLargestAtomicFloat               | None           |
| ProdIntDivRoundTo                    | Undefined      |
| ProdEndianess                        | Unspecified    |
| ProdWordSize                         | 32             |
| ProdShiftRightIntArith               | on             |
| ProdLongLongMode                     | off            |
| ProdHWDeviceType                     | 32-bit Generic |
| TargetBitPerChar                     | 8              |
| TargetBitPerShort                    | 16             |
| TargetBitPerInt                      | 32             |
| TargetBitPerLong                     | 32             |
| TargetBitPerLongLong                 | 64             |
| TargetBitPerFloat                    | 32             |
| TargetBitPerDouble                   | 64             |
| TargetBitPerPointer                  | 32             |
| TargetLargestAtomicInteger           | Char           |
| TargetLargestAtomicFloat             | None           |
| TargetShiftRightIntArith             | on             |
| TargetLongLongMode                   | off            |
| TargetIntDivRoundTo                  | Undefined      |
| TargetEndianess                      | Unspecified    |
| TargetWordSize                       | 32             |
| TargetTypeEmulationWarnSuppressLevel | 0              |
| TargetPreprocMaxBitsSint             | 32             |
| TargetPreprocMaxBitsUint             | 32             |
| TargetHWDeviceType                   | Specified      |
| TargetUnknown                        | off            |
| ProdEqTarget                         | on             |

**Table 6.7. model\_tugas3 Configuration Set.Components [29](6)**

| Property                                 | Value                         |
|------------------------------------------|-------------------------------|
| Name                                     | Model Referencing             |
| Description                              |                               |
| Components                               |                               |
| UpdateModelReferenceTargets              | IfOutOfDateOrStructuralChange |
| CheckModelReferenceTargetMessage         | error                         |
| EnableParallelModelReferenceBuilds       | off                           |
| ParallelModelReferenceErrorOnInvalidPool | on                            |
| ParallelModelReferenceMATLABWorkerInit   | None                          |
| ModelReferenceNumInstancesAllowed        | Multi                         |
| PropagateVarSize                         | Infer from blocks in model    |
| ModelDependencies                        |                               |
| ModelReferencePassRootInputsByReference  | on                            |
| ModelReferenceMinAlgLoopOccurrences      | off                           |
| PropagateSignalLabelsOutOfModel          | off                           |
| SupportModelReferenceSimTargetCustomCode | off                           |

**Table 6.8. model\_tugas3 Configuration Set.Components [29](7)**

| Property               | Value             |
|------------------------|-------------------|
| Name                   | Simulation Target |
| Description            |                   |
| Components             |                   |
| SimCustomSourceCode    |                   |
| SimCustomHeaderCode    |                   |
| SimCustomInitializer   |                   |
| SimCustomTerminator    |                   |
| SimReservedNameArray   |                   |
| SimUserSources         |                   |
| SimUserIncludeDirs     |                   |
| SimUserLibraries       |                   |
| SFSimEnableDebug       | on                |
| SFSimOverflowDetection | on                |
| SFSimEcho              | on                |
| SimBlas                | on                |
| SimCtrlC               | on                |
| SimExtrinsic           | on                |
| SimIntegrity           | on                |
| SimUseLocalCustomCode  | off               |

|                    |                      |
|--------------------|----------------------|
| SimParseCustomCode | on                   |
| SimBuildMode       | sf_incremental_build |
| SimDataInitializer |                      |

**Table 6.9. model\_tugas3 Configuration Set.Components [29](8)**

| Property                         | Value           |
|----------------------------------|-----------------|
| Name                             | Code Generation |
| SystemTargetFile                 | grt.tlc         |
| TLCOptions                       |                 |
| CodeGenDirectory                 |                 |
| GenCodeOnly                      | off             |
| MakeCommand                      | make_rtw        |
| GenerateMakefile                 | on              |
| PackageGeneratedCodeAndArtifacts | off             |
| PackageName                      |                 |
| TemplateMakefile                 | grt_default_tmf |
| PostCodeGenCommand               |                 |
| Description                      |                 |
| GenerateReport                   | off             |
| SaveLog                          | off             |
| RTWVerbose                       | on              |
| RetainRTWFile                    | off             |
| ProfileTLC                       | off             |
| TLCDebug                         | off             |
| TLCCoverage                      | off             |
| TLCAssert                        | off             |
| ProcessScriptMode                | Default         |
| ConfigurationMode                | Optimized       |
| ProcessScript                    |                 |
| ConfigurationScript              |                 |
| ConfigAtBuild                    | off             |
| RTWUseLocalCustomCode            | off             |
| RTWUseSimCustomCode              | off             |
| CustomSourceCode                 |                 |
| CustomHeaderCode                 |                 |
| CustomInclude                    |                 |
| CustomSource                     |                 |
| CustomLibrary                    |                 |
| CustomInitializer                |                 |

|                                                 |                                                                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| CustomTerminator                                |                                                                                                                                        |
| IncludeHyperlinkInReport                        | off                                                                                                                                    |
| LaunchReport                                    | off                                                                                                                                    |
| PortableWordSizes                               | off                                                                                                                                    |
| GenerateErtSFunction                            | off                                                                                                                                    |
| CreateSILPILBlock                               | None                                                                                                                                   |
| CodeExecutionProfiling                          | off                                                                                                                                    |
| CodeExecutionProfileVariable                    | executionProfile                                                                                                                       |
| CodeProfilingSaveOptions                        | SummaryOnly                                                                                                                            |
| CodeProfilingInstrumentation                    | off                                                                                                                                    |
| SILDebugging                                    | off                                                                                                                                    |
| TargetLang                                      | C                                                                                                                                      |
| IncludeRootSignalInRTWFile                      | off                                                                                                                                    |
| IncludeVirtualBlocksInRTWFileBlockHierarchy-Map | off                                                                                                                                    |
| IncludeRegionsInRTWFileBlockHierarchyMap        | off                                                                                                                                    |
| IncludeERTFirstTime                             | off                                                                                                                                    |
| GenerateTraceInfo                               | off                                                                                                                                    |
| GenerateTraceReport                             | off                                                                                                                                    |
| GenerateTraceReportSl                           | off                                                                                                                                    |
| GenerateTraceReportSf                           | off                                                                                                                                    |
| GenerateTraceReportEml                          | off                                                                                                                                    |
| GenerateCodeInfo                                | off                                                                                                                                    |
| GenerateWebview                                 | off                                                                                                                                    |
| GenerateCodeMetricsReport                       | off                                                                                                                                    |
| GenerateCodeReplacementReport                   | off                                                                                                                                    |
| RTWCompilerOptimization                         | Off                                                                                                                                    |
| ObjectivePriorities                             |                                                                                                                                        |
| RTWCustomCompilerOptimizations                  |                                                                                                                                        |
| CheckMdlBeforeBuild                             | Off                                                                                                                                    |
| CustomRebuildMode                               | OnUpdate                                                                                                                               |
| DataInitializer                                 |                                                                                                                                        |
| Components                                      | [ model_tugas3 Configuration Set.Components(8)-.Components(1) [39] , model_tugas3 Configuration Set.Components(8).Components(2) [41] ] |

**Table 6.10. model\_tugas3 Configuration Set.Components [29](9)**

| Property    | Value                                    |
|-------------|------------------------------------------|
| Description | HDL Coder custom configuration component |

|            |           |
|------------|-----------|
| Components |           |
| Name       | HDL Coder |

**Table 6.11. model\_tugas3 Configuration Set.Components [29](10)**

| Property                        | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Components                      | [ model_tugas3 Configuration Set.Components(-10).Components(1) [42] , model_tugas3 Configuration Set.Components(10).Components(2)-[43] ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Name                            | Simscape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| EditingMode                     | Full                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ExplicitSolverDiagnosticOptions | warning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| GlobalZcOffDiagnosticOptions    | warning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| SimscapeLogType                 | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| SimscapeLogSimulationStatistics | off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| SimscapeLogName                 | simlog                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| SimscapeLogDecimation           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| SimscapeLogLimitData            | on                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| SimscapeLogDataHistory          | 5000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| SelectedTab                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Version                         | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ComponentsAttached              | true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Listener                        | [ model_tugas3 Configuration Set.Components(1-10).Listener(1) [44] , model_tugas3 Configuration Set.Components(10).Listener(2) [44] , model_tugas3 Configuration Set.Components(10).Listener(3) [44] , model_tugas3 Configuration Set.Components(10).Listener(4) [44] , model_tugas3 Configuration Set.Components(10).Listener(5) [44] , model_tugas3 Configuration Set.Components(10).Listener(6) [44] , model_tugas3 Configuration Set.Components(10).Listener(7) [44] , model_tugas3 Configuration Set.Components(10).Listener(8) [44] , model_tugas3 Configuration Set.Components(10).Listener(9) [44] ] |
| someListenersNotInstalled       | false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| instanceId                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Table 6.12. model\_tugas3 Configuration Set.Components(8).Components [38](1)**

| Property | Value |
|----------|-------|
|----------|-------|

| Name                        | Code Appearance |
|-----------------------------|-----------------|
| Description                 |                 |
| Components                  |                 |
| Comment                     |                 |
| ForceParamTrailComments     | off             |
| GenerateComments            | on              |
| IgnoreCustomStorageClasses  | on              |
| IgnoreTestpoints            | off             |
| IncHierarchyInIds           | off             |
| MaxIdLength                 | 31              |
| PreserveName                | off             |
| PreserveNameWithParent      | off             |
| ShowEliminatedStatement     | off             |
| OperatorAnnotations         | off             |
| IncAutoGenComments          | off             |
| SimulinkDataObjDesc         | off             |
| SFDataObjDesc               | off             |
| MATLABFcnDesc               | off             |
| IncDataTypeInIds            | off             |
| PrefixModelToSubsysFcnNames | on              |
| MangleLength                | 1               |
| CustomSymbolStr             | \$R\$N\$M       |
| CustomSymbolStrGlobalVar    | \$R\$N\$M       |
| CustomSymbolStrType         | \$N\$R\$M_T     |
| CustomSymbolStrField        | \$N\$M          |
| CustomSymbolStrFcn          | \$R\$N\$M\$F    |
| CustomSymbolStrFcnArg       | rt\$I\$N\$M     |
| CustomSymbolStrBlkIO        | rtb_\$N\$M      |
| CustomSymbolStrTmpVar       | \$N\$M          |
| CustomSymbolStrMacro        | \$R\$N\$M       |
| CustomCommentsFcn           |                 |
| DefineNamingRule            | None            |
| DefineNamingFcn             |                 |
| ParamNamingRule             | None            |
| ParamNamingFcn              |                 |
| SignalNamingRule            | None            |
| SignalNamingFcn             |                 |
| InsertBlockDesc             | off             |
| InsertPolySpaceComments     | off             |

|                       |           |
|-----------------------|-----------|
| SimulinkBlockComments | on        |
| MATLABSourceComments  | off       |
| EnableCustomComments  | off       |
| InternalIdentifier    | Shortened |
| InlinedPrmAccess      | Literals  |
| ReqsInCode            | off       |
| UseSimReservedNames   | off       |
| ReservedNameArray     |           |

**Table 6.13. model\_tugas3 Configuration Set.Components(8).Components [38](2)**

| Property                         | Value                  |
|----------------------------------|------------------------|
| Name                             | Target                 |
| Description                      |                        |
| Components                       |                        |
| IsERTTarget                      | off                    |
| TargetFcnLib                     | ansi_tfl_table_tmw.mat |
| TargetLibSuffix                  |                        |
| TargetPreCompLibLocation         |                        |
| GenFloatMathFcnCalls             | ANSI_C                 |
| TargetFunctionLibrary            | ANSI_C                 |
| CodeReplacementLibrary           | ANSI_C                 |
| UtilityFuncGeneration            | Auto                   |
| ERTMultiwordTypeDef              | System defined         |
| ERTMultiwordLength               | 256                    |
| MultiwordLength                  | 2048                   |
| GenerateFullHeader               | on                     |
| GenerateSampleERTMain            | off                    |
| GenerateTestInterfaces           | off                    |
| IsPILTarget                      | off                    |
| ModelReferenceCompliant          | on                     |
| ParMdlRefBuildCompliant          | on                     |
| CompOptLevelCompliant            | on                     |
| ConcurrentExecutionCompliant     | on                     |
| IncludeMdlTerminateFcn           | on                     |
| GeneratePreprocessorConditionals | Disable all            |
| CombineOutputUpdateFcns          | on                     |
| CombineSignalStateStructs        | off                    |
| SuppressErrorStatus              | off                    |

|                                            |          |
|--------------------------------------------|----------|
| ERTFirstTimeCompliant                      | off      |
| IncludeFileDelimiter                       | Auto     |
| ERTCustomFileBanners                       | off      |
| SupportAbsoluteTime                        | on       |
| LogVarNameModifier                         | rt_      |
| MatFileLogging                             | on       |
| MultiInstanceERTCode                       | off      |
| SupportNonFinite                           | on       |
| SupportComplex                             | on       |
| PurelyIntegerCode                          | off      |
| SupportContinuousTime                      | on       |
| SupportNonInlinedSFcns                     | on       |
| SupportVariableSizeSignals                 | off      |
| ParenthesesLevel                           | Nominal  |
| GenerateClassInterface                     | off      |
| ModelStepFunctionPrototypeControlCompliant | off      |
| CPPClassGenCompliant                       | off      |
| AutosarCompliant                           | off      |
| GRTInterface                               | off      |
| GenerateAllocFcn                           | off      |
| ExtMode                                    | off      |
| ExtModeStaticAlloc                         | off      |
| ExtModeTesting                             | off      |
| ExtModeStaticAllocSize                     | 1000000  |
| ExtModeTransport                           | 0        |
| ExtModeMexFile                             | ext_comm |
| ExtModeMexArgs                             |          |
| ExtModelIntrflLevel                        | Level1   |
| RTWCAPISignals                             | off      |
| RTWCAPIParams                              | off      |
| RTWCAPISates                               | off      |
| RTWCAPIRootIO                              | off      |
| GenerateASAP2                              | off      |

**Table 6.14. model\_tugas3 Configuration Set.Components(10).Components [39](1)**

| Property    | Value |
|-------------|-------|
| Description |       |
| Components  |       |



| Name                          | SimMechanics                       |
|-------------------------------|------------------------------------|
| WarnOnRedundantConstraints    | on                                 |
| WarnOnSingularInitialAssembly | off                                |
| ShowCutJoints                 | off                                |
| VisOnUpdateDiagram            | off                                |
| VisDuringSimulation           | off                                |
| EnableVisSimulationTime       | on                                 |
| VisSampleTime                 | 0                                  |
| DisableBodyVisControl         | off                                |
| ShowCG                        | on                                 |
| ShowCS                        | on                                 |
| ShowOnlyPortCS                | off                                |
| HighlightModel                | on                                 |
| FramesToBeSkipped             | 0                                  |
| AnimationDelay                | 3                                  |
| RecordAVI                     | off                                |
| CompressAVI                   | on                                 |
| AviFileName                   |                                    |
| AutoFitVis                    | off                                |
| EnableSelection               | on                                 |
| LastVizWinPosition            | [-1 -1 -1 -1]                      |
| CamPosition                   | [0 0 0]                            |
| CamTarget                     | [0 0 -1]                           |
| CamUpVector                   | [0 1 0]                            |
| CamHeight                     | -1                                 |
| CamViewAngle                  | 0                                  |
| VisBackgroundColor            | [0.9 0.9 0.95]                     |
| DefaultBodyColor              | [1 0 0]                            |
| MDLBodyVisualizationType      | Convex hull from body CS locations |
| OVRRIDBodyVisualizationType   | NONE                               |
| VisConfigFile                 |                                    |

**Table 6.15. model\_tugas3 Configuration Set.Components(10).Components [39](2)**

| Property    | Value                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | SimMechanics 2G                                                                                                                                                      |
| Components  | [ model_tugas3 Configuration Set.Components(1-0).Components(2).Components(1) [44] , model_tugas3 Configuration Set.Components(10).Components(2).Components(2) [45] ] |

|      |                |
|------|----------------|
| Name | SimMechanics2G |
|------|----------------|

**model\_tugas3 Configuration Set.Components(10).Listener(1) (handle.listener, )**

Note: this object has no unfiltered properties.

**model\_tugas3 Configuration Set.Components(10).Listener(2) (handle.listener, )**

Note: this object has no unfiltered properties.

**model\_tugas3 Configuration Set.Components(10).Listener(3) (handle.listener, )**

Note: this object has no unfiltered properties.

**model\_tugas3 Configuration Set.Components(10).Listener(4) (handle.listener, )**

Note: this object has no unfiltered properties.

**model\_tugas3 Configuration Set.Components(10).Listener(5) (handle.listener, )**

Note: this object has no unfiltered properties.

**model\_tugas3 Configuration Set.Components(10).Listener(6) (handle.listener, )**

Note: this object has no unfiltered properties.

**model\_tugas3 Configuration Set.Components(10).Listener(7) (handle.listener, )**

Note: this object has no unfiltered properties.

**model\_tugas3 Configuration Set.Components(10).Listener(8) (handle.listener, )**

Note: this object has no unfiltered properties.

**model\_tugas3 Configuration Set.Components(10).Listener(9) (handle.listener, )**

Note: this object has no unfiltered properties.

**Table 6.16. model\_tugas3 Configuration Set.Components(10).Components(2).Components [43](1)**

| Property                                   | Value                |
|--------------------------------------------|----------------------|
| Description                                | Diagnostics          |
| Components                                 |                      |
| Name                                       | DiagnosticsConfigSet |
| SimMechanicsInvalidVisualProperty          | warning              |
| SimMechanicsCrossSectionNullEdge           | warning              |
| SimMechanicsUnconnectedFramePorts          | warning              |
| SimMechanicsRedundantBlock                 | warning              |
| SimMechanicsConflictingReferenceFrames     | warning              |
| SimMechanicsRigidlyBoundBlock              | error                |
| SimMechanicsUnsatisfiedHighPriorityTargets | warning              |

|                                          |       |
|------------------------------------------|-------|
| SimMechanicsJointTargetOverSpecification | error |
|------------------------------------------|-------|

**Table 6.17. model\_tugas3 Configuration Set.Components(10).Components(2).Components [43](2)**

| Property                             | Value             |
|--------------------------------------|-------------------|
| Description                          | Explorer          |
| Components                           |                   |
| Name                                 | ExplorerConfigSet |
| SimMechanicsOpenEditorOnUpdate       | on                |
| InternalSimMechanicsExplorerSettings |                   |

---

# Chapter 7. Glossary

**Atomic Subsystem.** A subsystem treated as a unit by an implementation of the design documented in this report. The implementation computes the outputs of all the blocks in the atomic subsystem before computing the next block in the parent system's block execution order (sorted list).

**Block Diagram.** A Simulink block diagram represents a set of simultaneous equations that relate a system or subsystem's inputs to its outputs as a function of time. Each block in the diagram represents an equation of the form  $y = f(t, x, u)$  where  $t$  is the current time,  $u$  is a block input,  $y$  is a block output, and  $x$  is a system state (see the Simulink documentation for information on the functions represented by the various types of blocks that make up the diagram). Lines connecting the blocks represent dependencies among the blocks, i.e., inputs whose current values are the outputs of other blocks. An implementation of a design described in this document computes a root or atomic system's outputs at each time step by computing the outputs of the blocks in an order determined by block input/output dependencies.

**Block Parameter.** A variable that determines the output of a block along with its inputs, for example, the gain parameter of a Gain block.

**Block Execution Order.** The order in which Simulink evaluates blocks during simulation of a model. The block execution order determined by Simulink ensures that a block executes only after all blocks on whose outputs it depends are executed.

**Checksum.** A number that indicates whether different versions of a model or atomic subsystem differ functionally or only cosmetically. Different checksums for different versions of the same model or subsystem indicate that the versions differ functionally.

**Design Variable.** A symbolic (MATLAB) variable or expression used as the value of a block parameter. Design variables allow the behavior of the model to be altered by altering the value of the design variable.

**Signal.** A block output, so-called because block outputs typically vary with time.

**Virtual Subsystem.** A subsystem that is purely graphical, i.e., is intended to reduce the visual complexity of the block diagram of which it is a subsystem. An implementation of the design treats the blocks in the subsystem as part of the first nonvirtual ancestor of the virtual subsystem (see Atomic Subsystem).

---

# Chapter 8. About this Report

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## 8.1. Report Overview

This report describes the design of the Simulation Analysis of Ohm's and Kirchhoff's Laws system. The report was generated automatically from a Simulink model used to validate the design. It contains the following sections:

**Model Version.** Specifies information about the version of the model from which this design description was generated. Includes the model checksum, a number that indicates whether different versions of the model differ functionally or only cosmetically. Different checksums for different versions indicate that the versions differ functionally.

**Root System.** Describes the design's root system.

**Subsystems.** Describes each of the design's subsystems.

**Design Variables.** Describes system design variables, i.e., MATLAB variables and expressions used as block parameter values.

**System Model Configuration.** Lists the configuration parameters, e.g., start and stop time, of the model used to simulate the system described by this report.

**Requirements Traceability.** Shows design requirements associated with elements of the design model. This section appears only if the design model contains requirements links.

**Glossary.** Defines Simulink terms used in this report.

## 8.2. Root System Description

This section describes a design's root system. It contains the following sections:

**Diagram.** Simulink block diagram that represents the algorithm used to compute the root system's outputs.

**Description.** Description of the root system. This section appears only if the model's root system has a Documentation property or a Doc block.

**Interface.** Name, data type, width, and other properties of the root system's input and output signals. The number of the block port that outputs the signal appears in angle brackets appended to the signal name. This section appears only if the root system has input or output ports.

**Blocks.** This section has two subsections:

- **Parameters.** Describes key parameters of blocks in the root system. This section also includes graphical and/or tabular representations of lookup table data used by lookup table blocks, i.e., blocks that use lookup tables to compute their outputs.
- **Block Execution Order.** Order in which blocks must be executed at each time step in order to ensure that each block's inputs are available when it executes.

**State Charts.** Describes state charts used in the root system. This section appears only if the root system contains Stateflow blocks.

## 8.3. Subsystem Descriptions

This section describes a design's subsystems. Each subsystem description contains the following sections:

**Checksum.** This section appears only if the subsystem is an atomic subsystem. The checksum indicates whether the version of the model subsystem used to generate this report differs functionally from other versions of the model subsystem. If two model checksums differ, the corresponding versions of the model differ functionally.

**Diagram.** Simulink block diagram that graphically represents the algorithm used to compute the subsystem's outputs.

**Description.** Description of the subsystem. This section appears only if the subsystem has a Documentation property or contains a Doc block.

**Interface.** Name, data type, width, and other properties of the subsystem's input and output signals. The number of the block port that outputs the signal appears in angle brackets appended to the signal name. This section appears only if the subsystem is atomic and has input or output ports.

**Blocks.** Blocks that this subsystem contains. This section has two subsections:

- **Parameters.** Key parameters of blocks in the subsystem. This section also includes graphical and/or tabular representations of lookup table data used by lookup table blocks, blocks that use lookup tables to compute their outputs.
- **Block Execution Order.** Order in which the subsystem's blocks must be executed at each time step in order to ensure that each block's inputs are available when the block executes. This section appears only if the subsystem is atomic.

**State Charts.** Describes state charts used in the subsystem. This section appears only if the root system contains Stateflow blocks.

## 8.4. State Chart Descriptions

This section describes the state machines used by Stateflow blocks to compute their outputs, i.e., Stateflow blocks. Each state machine description contains the following sections:

**Chart.** Diagram representing the state machine.

**States.** Describes the state machine's states. Each state description includes the state's diagram and diagrams and/or descriptions of graphical functions, Simulink functions, truth tables, and MATLAB functions parented by the state.

**Transitions.** Transitions between the state machine's states. Each transition description specifies the values of key transition properties. Appears only if a transition has properties that do not appear on the chart.

**Junctions.** Transition junctions. Each junction description specifies the values of key junction properties. Appears only if a junction has properties that do not appear on the chart.

**Events.** Events that trigger state transitions. Each event description specifies the values of key event properties.

**Data.** Data types and other properties of the Stateflow block's inputs, outputs, and other state machine data.

**Targets.** Executable implementations of the state machine used to compute the outputs of the corresponding Stateflow block.

**MATLAB Supporting Functions.** List of functions invoked by MATLAB functions defined in the chart.